



Data Science Team — Dataset Exercise for Candidates

Introduction

Thank you for taking the time to interview with us so far — we're excited about your interest in a role on our team and would like to invite you back for another discussion. In advance of our next meeting, we'd like to ask you to tackle the below real-world business problem (for a fictional new “buy now, pay later” fintech company called “Spendy”) so we can see how you work. We'll ask you to make a small, informal presentation of your work to members of the Data Science Team at our next meeting.

Before diving into the problem, please go ahead and schedule your next interview with our recruiter. We prefer that you set a time approximately one week after receiving these instructions, but the recruiter will work with you one-on-one to arrange a time that is convenient for everyone.

As you will see, the below exercise is open-ended enough that in a real business setting it could be a month-long project (and there would be more data to work with!). However, we would like you to approach this in a “hack-a-thon” style – getting as far along as you can in a fixed amount of time – and will evaluate your result and presentation based on that expectation. Please do not spend more than a day working on the assignment.

Instructions

1. Like many new online services, as Spendy grows, customer loyalty becomes just as important as new customer acquisition. But what is a loyal customer for Spendy? The head of marketing has tasked you with analyzing patterns in Spendy customer purchase data to propose a definition of a “loyal customer” and make it possible to count and track how many loyal customers there are over time. Make sure to show the analysis you did to support your proposal.
 - Some points to get you thinking in the right direction for the analysis:
 - Consider from the customer's perspective the difference between loyalty to a specific partner shop and loyalty to Spendy.
 - There may be some cyclical and/or seasonal trends to analyze..
 - Both the purchase frequency of a customer as well as their spending amount should be considered as important factors for defining customer loyalty.
2. The head of marketing also wants to know if it's possible to predict from a brand new customer's first two months of activity whether or not that user will become a loyal customer (under the

definition you proposed). They want to be able to segment customers by the likelihood they'll become loyal and then develop different communications strategies for each segment. Can you propose an approach using machine learning to model the likelihood a new customer will become a loyal customer?

- Some guidelines:
 - Make sure to show how you transform the raw customer purchase data into training data that works for the business problem.
 - Please calculate a couple simple features from the raw customer purchase data that could be useful for predicting customer loyalty.
 - Using the training data, train a model for this prediction problem and show how you evaluate the performance of the model.
 - Our evaluation will focus on your approach and setup more than your final result and model performance. There is no need to write production-ready code nor do any hyperparameter tuning or other efforts to boost your model's performance (remember this is a “hack-a-thon” style assignment!).

Deliverables

- Please present your findings, (a typical presentation is about 30 minutes plus time for questions/discussion). The audience will mostly be other data scientists (members of the team you're applying to join), but we may also invite some guest attendees from data engineering and marketing. You can focus your presentation to an audience of data scientists as if it's the team internal review before sharing with the head of marketing.
 - You can use whatever you'd like to present: markdown in a Jupyter notebook, PPT, etc. We don't want you to spend more than a day on this in total, so do not worry about making a nice-looking polished presentation document; we're going to be focused on your thinking and findings.
- Before the day of presentation (one day before is fine), please share your code to the recruiter or Paidy's HR so they can pass it to the Data Science team to understand your technical approach to arriving at your findings.
 - Please share your analysis/code via email rather than publishing it on Github or elsewhere.

Notes

3. The dataset you have available to work on (1) and (2) is a slice of transaction data from Spendy, logging when users use Spendy's “buy now, pay later” payment method to make a purchase at one of Spendy's partner merchants.
 - The columns of the dataset:

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- purchase_id: an ID representing the purchase
 - user_id: an ID representing the user (customer)
 - shop_id: an ID representing the partner merchant the user is making the purchase at
 - timestamp: the time that the purchase attempt is made
 - price: the amount the user spent
 - first_purchase: a boolean flag of whether this is the user's first ever purchase or not (if the value is False for all purchases from a user_id, it means they are an existing Spendy user who has made a purchase before the time range of the dataset)
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- It is important that you provide explanations of your work (i.e. why you choose to do things with the data set that you do). You are welcome to present verbally during the interview time, but your output should be a complete representation of your analysis
 - If you have any questions whatsoever, please feel free to reach out to us directly or via the recruiter!